



Low cycle fatigue of irradiated and unirradiated Eurofer97 steel at 300 °C

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ABSTRACT

Low cycle fatigue results are reported for unirradiated and irradiated reduced activation ferritic martensitic steel Eurofer97. The neutron irradiation experiment (irradiation at 300 °C to a nominal dose of 2.5 dpa) has been performed in the High Flux Reactor, Petten, the Netherlands. Post-irradiation low cycle fatigue tests have been performed in air at 300 °C at a total strain range of 0.6%, 1.0% and 1.4%. Neutron irradiation at 300 °C resulting in irradiation hardening is found to be beneficial for fatigue life at low strain amplitudes and to be adverse at high strain amplitudes. No effect of the different technological product forms on the fatigue life in Eurofer97 is observed, and fatigue behavior of Eurofer97 steel is found to be similar to that of F82H steel.

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1. Introduction

Reduced activation ferritic martensitic steel (RAFM), Eurofer97, is considered as one of the potential candidates for the structural components of the EU Test Blanket Module (TBM) of ITER reactor and of the future fusion power plants. In the current ITER TBM designs the TBM steel box will be manufactured from Eurofer97 steel [1–3]. The component will operate under combined mechanical and thermal cycling, thus in order to design it the low cycle fatigue life (LCF) has to be carefully examined and the effects of irradiation on the fatigue properties of Eurofer97 steel have to be understood. More detailed information on the operational conditions of the Test Blanket Module can be found elsewhere [1–3].

In the present work the low cycle fatigue life results for different technological products (14 mm plate, powder and solid HIP (Hot Isostatic Press)) of Eurofer97 steel are analyzed and discussed with respect to the irradiation effects and are compared with those for a F82H reduced activation ferritic martensitic steel.

2. Material and methods

The material used in this study is Eurofer97-1 (heat E83698) with a nominal composition of Fe–0.1C–9Cr–1W–0.2V–0.1Ta, wt.%. The material was produced by Böhler and delivered in the form of 14 mm plate. The final heat treatment was (1) austenitisation for 30 min at 980 °C followed by air cooling and (2) tempering for 90 min at 760 °C followed by air cooling.

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Several technological products of Eurofer97 were produced by CEA Grenoble. The 25 mm plate (heat E83697) of Eurofer97 has been used to produce two stacks of seven discs each of 70 mm diameter, which were surface treated and subsequently diffusion welded by CEA Grenoble (solid HIP). HIP was performed at 1100 °C. The post-HIP heat treatment was 60 min at 980 °C followed by 120 min at 750 °C. CEA Grenoble has also manufactured a batch of Eurofer97 powder from the 100 mm forged bar (heat E83699). The powder was subsequently HIPed to a solid block (powder HIP). The HIP cycle was for 240 min at 1040 °C. The post-HIP heat treatment was 60 min at 980 °C followed by 120 min at 750 °C.

In order to study the effects of irradiation damage on fatigue life of Eurofer97 a part of LCF specimens have been irradiated in sodium environment at 300 °C to a nominal dose of 2.5 dpa. The neutron irradiation experiment has been performed in the High Flux Reactor, Petten, the Netherlands.

Low cycle fatigue tests are performed in accordance with ASTM E606-92 standard using cylindrical specimens with a diameter of 3 mm and a total length of 45 mm. All tests are carried out in air at 300 °C. No hold time has been applied as creep–fatigue interaction effects are not expected at 300 °C for Eurofer97 steel. The applied total strain ranges 0.6%, 1.0% and 1.4% are chosen for the different values of elastic and plastic strain components. All tests are conducted until complete specimen separation; however the number of cycles reported is for the criterion of 50% stress reduction of the first cycle stress. It should be noted that at 50% stress decrease a deep crack in the specimen is already present and the specimen has been already failed in a technological sense.

3. Results and discussion

Fig. 1 shows the LCF results obtained for unirradiated Eurofer97 plate and HIPed materials (solid and powder HIP) at 300 °C. It can

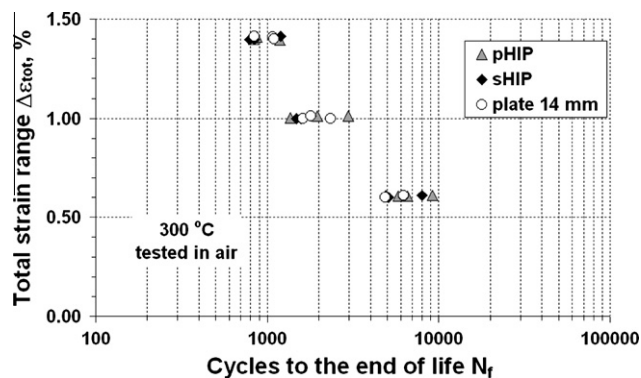


Fig. 1. LCF results for unirradiated Eurofer97 plate and HIPed (solid (sHIP) and powder (pHIP)) materials tested at 300 °C in air.

be seen that there is no significant difference in fatigue life for the different technological product forms.

It has been shown previously [4,5] that irradiation of Eurofer97 at 300 °C results in a strong irradiation hardening effect, the yield stress increase is about 300 MPa at 300 °C test temperature at a nominal dose of 2.5 dpa. As for the low cycle fatigue life of Eurofer97 it is obvious (Fig. 2) that the effect of irradiation hardening on LCF life is very different for the high and low strain range independent of the technological product form. It should be noted that the fatigue behavior of Eurofer97 steel is found to be similar to that of F82H steel [6]. A considerable scatter in fatigue life is observed, which is often attributed to the fact that fatigue is a stochastic process.

3.1. High strain range

It has been shown (Fig. 2) that at the high strain range irradiation has a detrimental effect on the low cycle fatigue life of Eurofer97 steel. In the high strain range the specimen is strained beyond the yield point. Fig. 3 shows an example of the first three cycles of stress–strain hysteresis loops for unirradiated (Fig. 3 a) and irradiated (Fig. 3 b) specimens at a total strain range ($\Delta\epsilon_{\text{tot}}$) of 1.4%, where the plastic tensile strain in the first cycle is more than 50% of the 0.2% yield point.

For the strongly irradiation hardened materials the deformation is immediately localized, supposedly creating softened channels clear of irradiation defects. In Eurofer97 steel no work hardening is observed [4] during tensile deformation, which can also be seen in Fig. 3 b. Once the material starts to yield, irradiation defects are

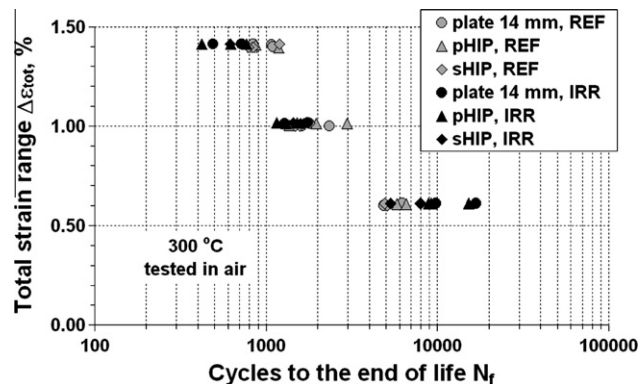


Fig. 2. LCF results for irradiated and unirradiated Eurofer97 plate and HIPed (solid (sHIP) and powder (pHIP)) material tested at 300 °C in air. REF – unirradiated material, IRR – irradiated material.

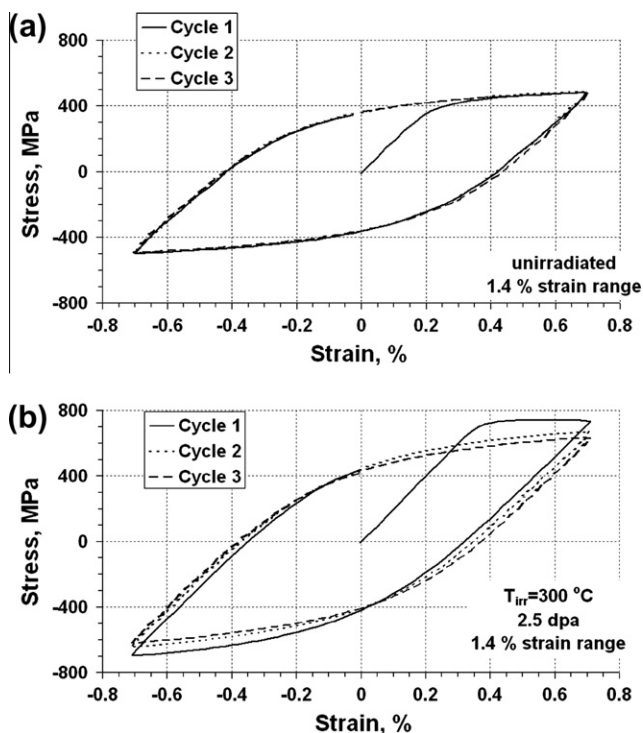


Fig. 3. Stress–strain hysteresis loops for first three cycles for (a) unirradiated and (b) irradiated Eurofer97 plate tested at 300 °C and 1.4% strain range.

being annihilated from the localized channels as the material deforms. Since the material surrounding these localized channels is harder than the material in the channels, all subsequent deformation will be concentrated in the softer channels. Thus more and more irradiation defects will be swept away leading to the rapid strain softening as the fatigue test progresses (Fig. 4). Eventually, all plastic deformation is concentrated in small regions of the material creating high local strains, and fatigue damage is building up faster than for softer (unirradiated) material, even if the total plastic strain is less. In unirradiated material, any localized deformation possibly leads to locally stronger material due to work hardening and further deformation occurs in the softer surrounding material.

3.2. Low strain range

At low strain ranges the irradiation hardening is actually beneficial for the fatigue life. At $\Delta\epsilon_{\text{tot}} < 0.8\%$ the yield point is not

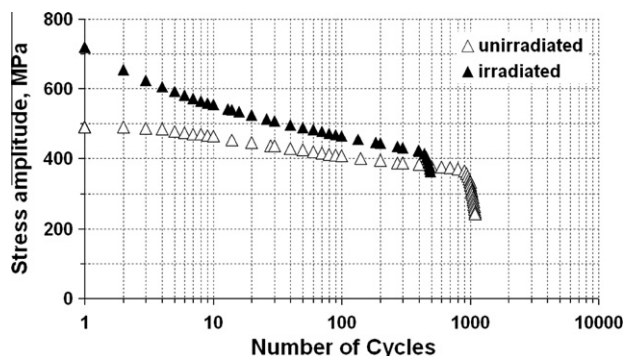


Fig. 4. Stress amplitude versus number of cycles for irradiated and unirradiated Eurofer97 plate tested at 300 °C and 1.4% strain range.

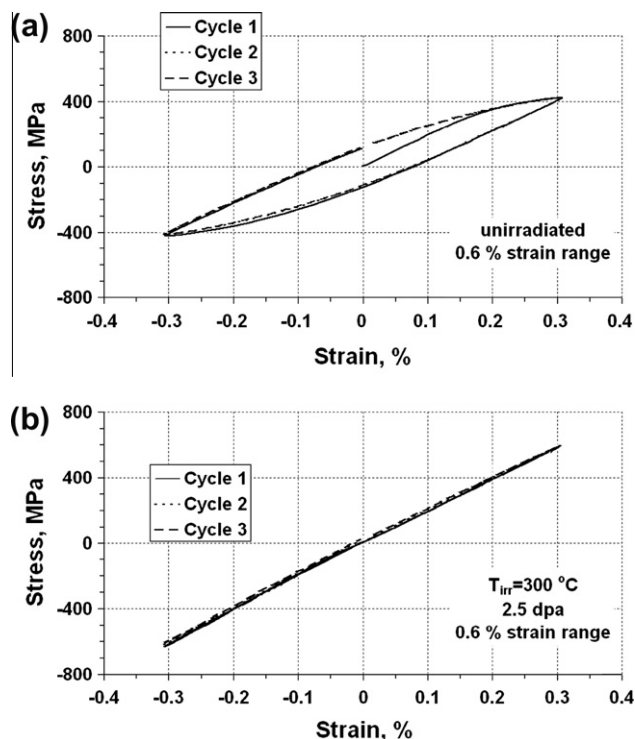


Fig. 5. Stress–strain hysteresis loops for first three cycles for (a) unirradiated and (b) irradiated Eurofer97 plate tested at 300 °C and 0.6% strain range.

reached yet, since the dislocations are much more strongly pinned by the irradiation defects and remain so for longer time than in the case of lower strength material. Fig. 5 shows an example of the first three cycles of stress–strain hysteresis loops for unirradiated (Fig. 5 a) and irradiated (Fig. 5 b) specimens at a total strain range of 0.6%. For irradiated material in the first cycle the plastic tensile strain is 0.01%, a factor of 20 below the plastic strain of 0.2% used for the engineering definition of the yield stress. No hysteresis is recorded for the first cycles for irradiated material (Fig. 5 b). This is due to the irradiation induced lattice damage. Fatigue damage in the form of dislocation cell formation and micro-crack initiation requires dislocation movement and it is postponed by the strong pinning obstacles induced by irradiation.

It is clear (Fig. 5 a) that for unirradiated specimen the plastic energy is stored in the material from the very beginning of the test. For unirradiated material in the first cycle the plastic tensile strain is 0.15%. During the entire fatigue life the plastic strain range for unirradiated material is larger than that for irradiated material;

therefore, it is not surprising that the fatigue life of the irradiated Eurofer97 steel increases compared with unirradiated material.

It is worth mentioning that it is always difficult to relate post-irradiation results with a real situation when the fatigue loading occurs simultaneously with the irradiation. It can be assumed that fatigue will partially annihilate the irradiation lattice damage preventing hardening to become significant. However, in the case of in-pile fatigue, the interaction between the formation of irradiation defects and concurrent dislocation movement may decrease the effects discussed above, or give rise to new damage mechanisms. But since the post-irradiation effect is not large, the interaction is not expected to be very dramatic.

4. Conclusions

The fatigue behavior of Eurofer97 steel is similar to that of the conventional F82H steel. A considerable scatter in fatigue life is observed in both steels, which is often attributed to the fact that fatigue is a stochastic process.

No pronounced effect of the different technological product forms (14 mm plate, solid and powder HIP) on the fatigue life in Eurofer97 has been observed.

Irradiation at low temperature (300 °C), when irradiation hardening occurs, is found to be beneficial for LCF life at low strain amplitudes, whereas it is adverse at high strain amplitudes. The possible explanation for such behavior is that in the low strain range the fatigue damage is postponed by the strong pinning obstacles induced by irradiation; and in the high strain range the fatigue damage is building up faster than for softer material due to localization of deformation. In order to confirm these observations an extensive investigation of the microstructure in irradiated Eurofer97 is necessary.

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